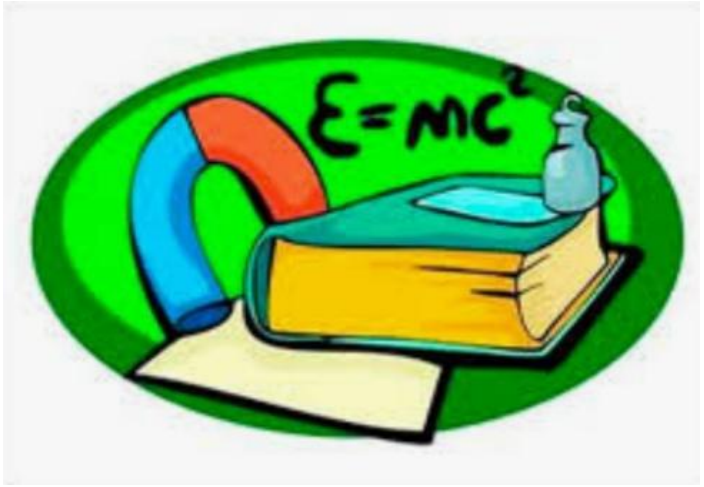
Fizika

9-sinf

Mavzu: **Umumiy takrorlash darsi**



Issiqlik miqdori va issiqlik balans tenglamasiga doir masala

22 ℃ haroratni ko‘rsatib turgan termometr suvga tushirilgandan so‘ng 70℃ ni ko‘rsatadi.

Termometr tushirilgunga qadar suvning harorati (℃) qanchaga teng bo‘lgan?

Suvning massasi 40 g, suvning solishtirma issiqlik sig‘imi 4200 $\frac{J}{kg\cdot℃}$, Termometrning issiqlik sig‘imi esa 7 $\frac{J}{℃}$.

Berilgan:

$t_1 = 22℃$

$t_m = 70℃$

$m_2 = 40\ g = 0{,}04\ kg$

$c_2 = 4200\ \frac{J}{kg \cdot ℃}$

$C_1 = 7\ \frac{J}{℃}$

Topish kerak: $t_2 - ?$

Formula:

$$Q_1 + Q_2 = 0 \Rightarrow \quad Q_1 = -Q_2$$

$$Q_1 = C_1 \Delta t_1 = C_1 (t_m - t_1)$$

$$Q_2 = c_2 m_2 \Delta t_2 = c_2 m_2 (t_m - t_2)$$

$$C_1 (t_m - t_1) = -c_2 m_2 (t_m - t_2)$$

$$C_1 (t_m - t_1) = -c_2 m_2 t_m + c_2 m_2 t_2$$

$$C_1 (t_m - t_1) + c_2 m_2 t_m = c_2 m_2 t_2$$

$$t_2 = \frac{C_1 (t_m - t_1) + c_2 m_2 t_m}{c_2 m_2}$$

$$t_2 = \frac{C_1(t_m - t_1) + c_2 m_2 t_m}{c_2 m_2}$$

Yechish:

$$t_2 = \frac{7\frac{J}{℃}\cdot(70℃-22℃)+4200\frac{J}{kg\cdot℃}\cdot 0{,}04\ kg\cdot 70℃}{4200\frac{J}{kg\cdot℃}\cdot 0{,}04\ kg} = 72℃$$

Javob: $t_2 = 72℃$

Issiqlik miqdori va foydali ish koeffitsiyentiga doir masala

600 W quvvatli elektroplitkada 3 l suvni qaynagunga qadar 40 minut isitiladi. Suvning boshlang‘ich temperaturasi $20℃$. Suvning solishtirma issiqlik sig‘imi $4200\frac{J}{kg℃}$.

Qurilmaning FIK aniqlang.

Berilgan:

$N = 600\ W$

$V = 3\ l = 0{,}003\ m^3$

$\tau = 40\ min = 2400\ s$

$t_1 = 20℃$

$\rho = 1000\ \frac{kg}{m^3}$

$c = 4200\ \frac{J}{kg℃}$

$t_q = 100℃$

Topish kerak: $\eta - ?$

Formula:

$$A_t = N\tau \qquad A_f = Q = cm\Delta t$$

$$\Delta t = t_q - t_1$$

$$\eta = \frac{A_f}{A_t} \cdot 100\ \% = \frac{cm(t_q - t_1)}{N\tau} \cdot 100\ \%$$

$$m = \rho V$$

$$\eta = \frac{c\rho V(t_q - t_1)}{N\tau} \cdot 100\ \%$$

$$\eta = \frac{c\rho V\left(t_q - t_1\right)}{N\tau} \cdot 100\ \%$$

Yechish:

$$\eta = \frac{4200\,\frac{J}{kg°\text{C}} \cdot 1000\,\frac{kg}{m^3} \cdot 0{,}003\ m^3 \cdot (100°\text{C} - 20°\text{C})}{600\ W \cdot 2400\ s} \cdot 100\ \% =$$

$$= 70\ \%$$

Javob: $\eta = 70\ \%$

Yoqilg‘ining solishtirma yonish issiqligi va issiqlik mashinalariga doir masala.

FIK 20 % bo‘lgan reaktiv samolyotning dvigateli 1800 km/h tezlik bilan uchishda 86 kN tortishish kuchiga erishadi.

Parvozning 1 soatdagi kerosin sarfini (tonnada) aniqlang. Kerosinning yonish issiqligi 43 MJ/kg.

Berilgan:

$$\vartheta = 1800\,\frac{km}{h} = 500\,\frac{m}{s}$$

$$t = 1\,h = 3600\,s$$

$$F = 86\,kN = 86000\,N$$

$$\eta = 20\,\%$$

$$q = 43\,\frac{MJ}{kg} = 43 \cdot 10^6\,\frac{J}{kg}$$

Topish kerak: $m - ?$

Formula:

$$A_t = qm\,;\; A_f = N_f t;\; \eta = \frac{A_f}{A_t} \cdot 100\,\%$$

$$\eta = \frac{N_f t}{qm} \cdot 100\,\% \Rightarrow m = \frac{N_f t \cdot 100\,\%}{\eta q}$$

$$N_f = F\vartheta$$

$$m = \frac{F\vartheta t \cdot 100\,\%}{\eta q}$$

Yechish:

$$m = \frac{86000\,N \cdot 500\frac{m}{s} \cdot 3600\,s \cdot 100\,\%}{20\,\% \cdot 43 \cdot 10^6 \frac{J}{kg}} = 18000\,kg = 18\,t$$

Javob: $m = 18\,t$

Termodinamikaning 1-qonuniga doir masala

Agar 0,5 mol gazga 290 J issiqlik miqdori berilib, haroratini izobar ravishda $27℃$ dan $47℃$ gacha oshirilsa, uning ichki energiyasi qanchaga o‘zgaradi?

Universal gaz doimiysi $8{,}3\ \frac{J}{℃\cdot mol}$.

Berilgan:

$\nu = 0{,}5\ mol$

$p = const.$

$Q = 290\ J$

$t_1 = 27°C$

$t_2 = 47°C$

$R = 8{,}3\ \frac{J}{°C \cdot mol}$

Formula:

$$Q = \Delta U + A$$

$$\Delta U = Q - A$$

$$A = \nu R \Delta T$$

$$\Delta T = \Delta t = t_2 - t_1$$

$$A = \nu R (t_2 - t_1)$$

$$\Delta U = Q - \nu R (t_2 - t_1)$$

Topish kerak: $\Delta U - ?$

Yechish: $\Delta U = 290\ J - 0{,}5\ mol \cdot 8{,}3\ \frac{J}{°C \cdot mol} \cdot (47°C - 27°C) = 207\ J$

Javob: $\Delta U = 207$ J

Ichki yonuv dvigatellariga doir masala

Ideal issiqlik mashinasi isitkichdan olingan issiqlikning 80% ini sovitkichga uzatadi.

Agar sovitkichning harorati 248 K bo‘lsa, isitkichning haroratini (K) toping.

Berilgan:

$100\ \% - \eta = 80\ \%$

$T_2 = 248\ K$

Topish kerak: $T_1 - ?$

Formula:

$$\eta = \frac{T_1 - T_2}{T_1} \cdot 100\ \%$$

$$\eta T_1 = T_1 \cdot 100\ \% - T_2 \cdot 100\ \%$$

$$T_2 \cdot 100\ \% = T_1 \cdot 100\ \% - \eta T_1$$

$$T_1 = \frac{T_2 \cdot 100\ \%}{100\ \% - \eta}$$

Yechish: $\eta = 100\ \% - 80\ \% = 20\ \%$

$$T_1 = \frac{248\ K \cdot 100\ \%}{100\ \% - 20\ \%} = 310\ K$$

Javob: $\boldsymbol{T_1 = 310\ K}$

Mustaqil bajarish uchun topshiriqlar

1. 54 km/h tezlikda 69 kW quvvatga erishadigan va FIK 40% bo‘lgan avtomobilning dvigateli uchun 10 kg benzin necha km yo‘lga yetadi? Benzinning solishtirma yonish issiqligi 46 MJ/kg.
2. 200 l sig‘imli vannani to‘ldirish uchun 10℃ li sovuq suvni 60℃ li issiq suv bilan aralashtirildi. Vannada 40℃ haroratni o‘rnatish uchun necha l sovuq suv solish kerak?

Mustaqil bajarish uchun topshiriqlar

3. 0,1 MPa bosimda $6\,l$ hajmni egallaydigan gaz silindrning porsheni ostida 300 K haroratda turadi. Gazni siqishda 50 J ga teng ish bajarilishi uchun uni o‘zgarmas bosimda necha gradusga sovitish kerak.
4. Ideal gaz Karno siklini bajaradi. Isitkichning absolyut harorati sovitkichning absolyut haroratidan 4 marta katta. Sovitkichga beriladigan issiqlik ulushini (foizlarda) aniqlang.